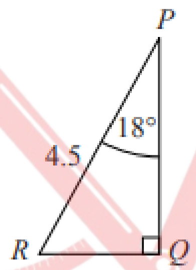
Trigonometry

Worksheet 1b

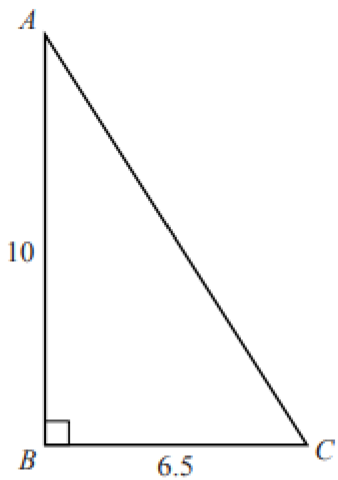
Basic, 90 degree only

1. **W12/qp21/q1(a)**



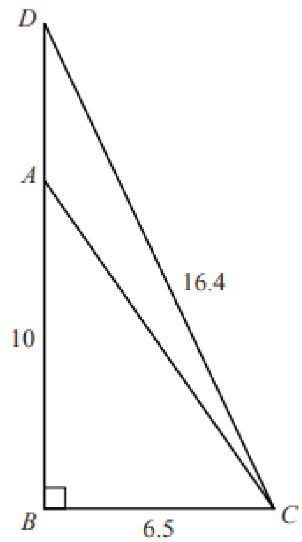
In triangle PQR, $R\hat{P}Q$ = 18°, $P\hat{Q}R$= 90° and PR = 4.5 m. Find PQ. [2]

2. **W12/qp22/q1(a)**



In triangle ABC, AB = 10 m, BC = 6.5 m and $A\hat{B}C$ = 90°.Find $A\hat{C}B$. [2]

3. W12/qp22/q1(b)

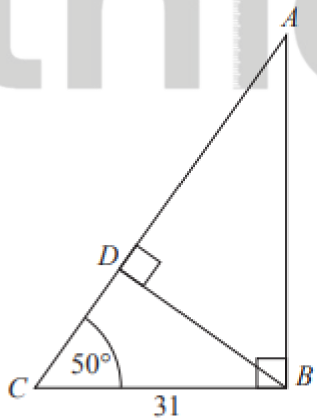


D is the point on BA produced such that CD = 16.4 m.

i. Find AD. Give your answer in metres and centimetres, correct to the nearest centimetre. [3]

ii. Find $D\hat{C}B$. [2]

4. W13/qp21/q6(a)

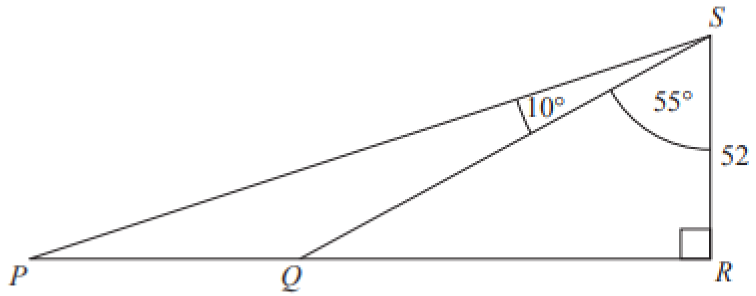


In the triangle ABC, $A\hat{B}C$ = 90°, $A\hat{C}B$ = 50° and BC = 31m. D is the point on AC such that $B\widehat{D}A$ = 90° .

i. Show that CD = 19.93m, correct to 2 decimal places. [2]

ii. Calculate AD. [3]

5. W13/qp21/q6(b)

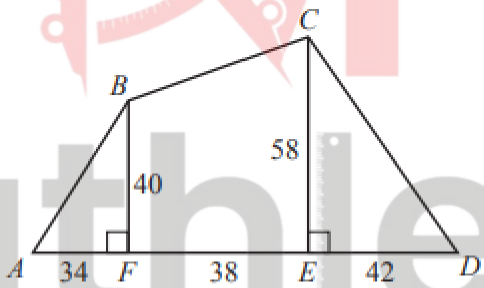


Two boats are at the points P and Q. RS is a vertical cliff of height 52m. $P\hat{S}Q$ = 10 and $Q\hat{R}S$= 55.

i. State the angle of depression of P from S. [1]

ii. Calculate the distance, PQ, between the boats. [3]

6. W13/qp22/q1



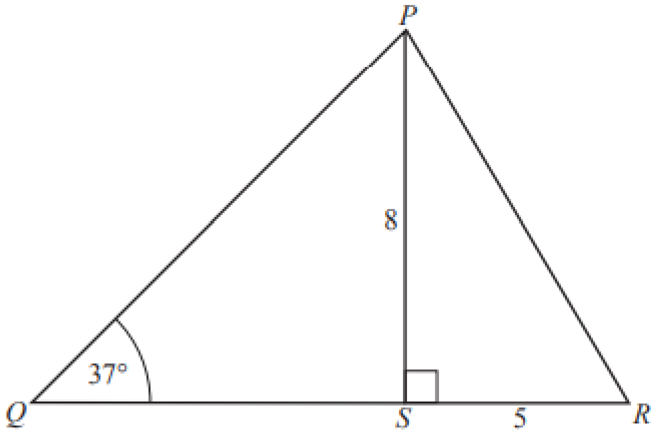
$ABCD$ is a level field. F and E are points on AD such that BF and CE are perpendicular to AD. BF = 40m and CE = 58m. AF = 34m, FE = 38m and ED = 42m.

a. Calculate the area of the field. [3]

b. Calculate the length of BC. [2]

c. Calculate $C\widehat{D}E$. [2]

7. W14/qp21/q8(b)

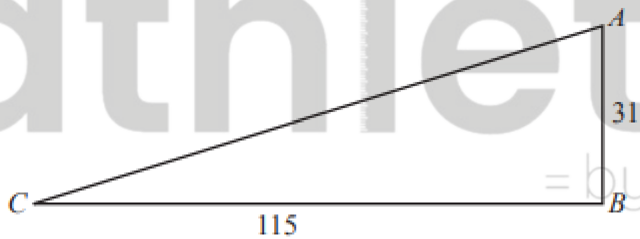


The diagram shows a triangle PQR with $P\hat{Q}R$ = 37°. S is the point on QR such that $P\hat{S}R$ = 90°, PS = 8 cm and SR = 5 cm. Calculate

i. PR, [2]

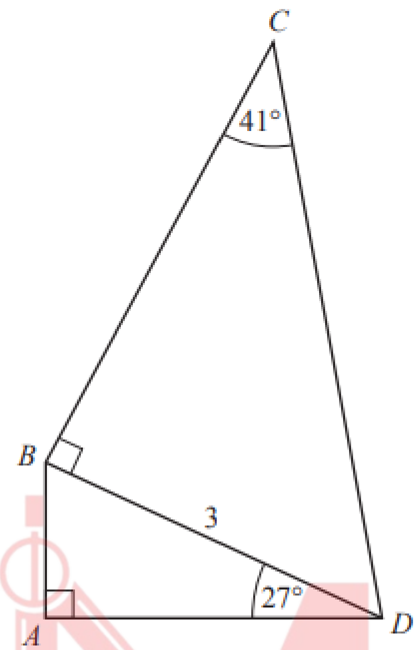
ii. the shortest distance from S to PQ. [3]

8. S16/qp21/q5(a)



AB is vertical and CB is horizontal. AB = 31m and CB = 115m. Calculate the angle of depression of C from A. [3]

9. W16/qp22/q4(a)



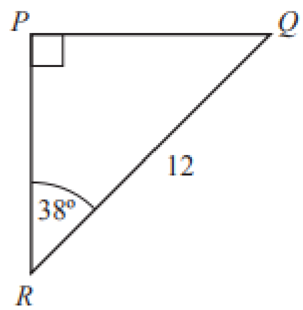
In the framework $ABCD$, BD = 3m.

$B\hat{D}A$ = 27°, $B\hat{C}D$ = 41°. $D\hat{B}C$ and $D\hat{A}B$ are right angles.

iii. Find AD. [2]

iv. Find CD. [3]

10. S17/qp22/q6(a)



Triangle PQR has a right angle at P, angle $P\hat{R}Q$ =38° and RQ=12 m. Calculate PQ.

[2

Answer Key

1. W12/qp21/q1(a)

1	(a) 4.28	2	**M1** for PQ = 4.5cos18 oe

2. W12/qp22/q1(a)

1	(a) 57(.0°)	2	**M1** for $\tan A\hat{C}B = \frac{10}{6.5}$ oe

3. W12/qp22/q1(b)

(b) (i) 5 m 6 cm cao	3	**B2** for ($BD =$) 15.1 or better or **M1** for $BD^2 = 16.4^2 - 6.5^2$ and/or **SC1** for their $BD - 10$
(ii) 66.6 or 66.7 (°)	2ft	e.g. accept $\tan^{-1}\frac{their\ DB}{6.5}$ **M1** for $\cos D\hat{C}B = \frac{6.5}{16.4}$ oe

4. W13/qp21/q6(a)

6	(a) (i) 19.93 from correct rounding	2	**M1** for $\frac{CD}{31} = \cos 50$ oe
	(ii) 28.3	3	**M1** for $\frac{31}{AC} = \cos 50$ oe and **M1** for $AC - 19.93$ **SC** If 2nd **M** not earned, **A1** for 48.2

5. W13/qp21/q6(b)

(b) (i) 25	1	
(ii) 37.2 or 37.3	3	**M1** for $\frac{PR}{52} = \tan 65$ oe or $\frac{QR}{52} = \tan 55$ oe and **M1** for $PR - QR$ **SC** If 2nd **M** not scored, **A1** for 111.5 or 74.26

6. W13/qp22/q1

1	(a) 3760	3	**B1** for a correct Δ such as $\frac{1}{2} \times 34 \times 40$ **B1** for $\frac{1}{2}$ $(40 + 58) \times 38$ oe soi
	(b) 42(.0)	2	**M1** for $(BC^2 =)\ 38^2 + (58 - 40)^2$
	(c) 54.1	2	**M1** for $\tan CDE = \frac{58}{42}$ oe

7. W14/qp21/q8(b)

(b) (i)	9.43	2	**M1** for $(PR^2 =)\ 5^2 + 8^2$
(ii)	6.38 to 6.39	3	**M2** for $\sin 53 = \frac{x}{8}$ oe or **B1** for correct triangle soi

8. S16/qp21/q5(a)

5 (a)	15.1 or 15.08(…..	3	M1 for $\tan\theta = \frac{31}{115}$ or $\tan\theta = \frac{115}{31}$ A1 for $\theta = 15.1$ or $\theta = 74.9$

9. W16/qp22/q4(a)

4 (a) (i)	2.67	2	M1 $\frac{AD}{3} = \cos 27$ oe
(ii)	4.57	3	M2 for $CD = \frac{3}{\sin 41}$ oe or M1 for $\frac{3}{CD} = \sin 41$ oe

10. S17/qp22/q6(a)

6(a)	7.387 to 7.392	2	**M1** for $\sin 38 = \frac{PQ}{12}$ soi or $\frac{PQ}{\sin 38} = \frac{12}{\sin 90}$ soi